

	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	15 Jul 2015	First Release	CHIN CHEE YAN
01	16 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

Content

Y Axis Travel Stroke Checking.....	3
Y Axis Optimization.....	7
Home sensor adjustment.....	7
Limit sensor adjustment.....	10

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

Y Axis Travel Stroke Checking

- Go to Digital I/O tab and initialize as show in Figure 1.

Services → Subsystem Status & Control → Digital I/O → Initialize

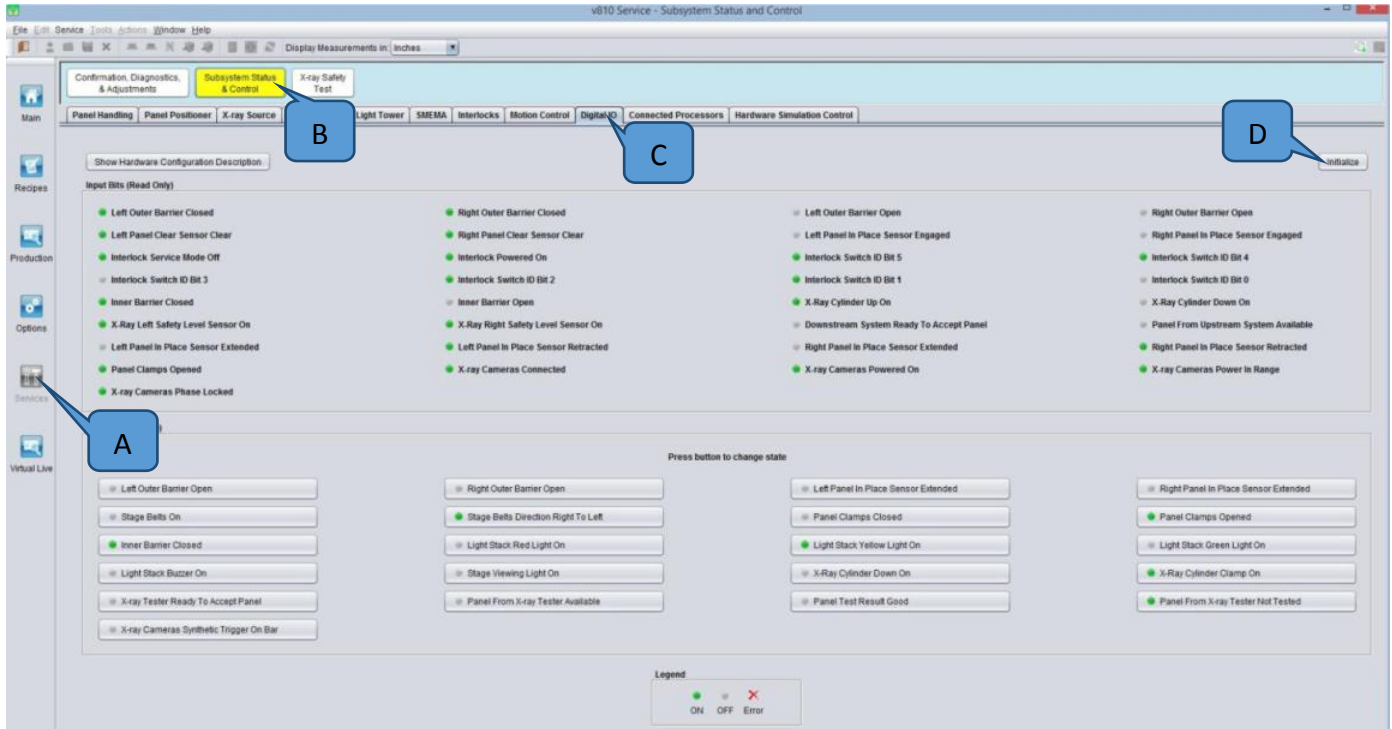


Figure 1 Digital IO

- Go to Panel Handling tab and initialize as show in Figure 5. Change display measurement unit to MM

Services → Subsystem Status & Control → Panel Handling → display unit to MM → Initialize



Figure 2 Panel Handling

	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

- Go to Panel Positional tab, click initialize and disable the Motion X&Y Axis. Turn off X ray and Open the front access door.

Services → Subsystem Status & Control → Panel Positional → Initialize → Disable X and Y Motion → Turn off X-ray Source Operator Console → Open Front Access Door

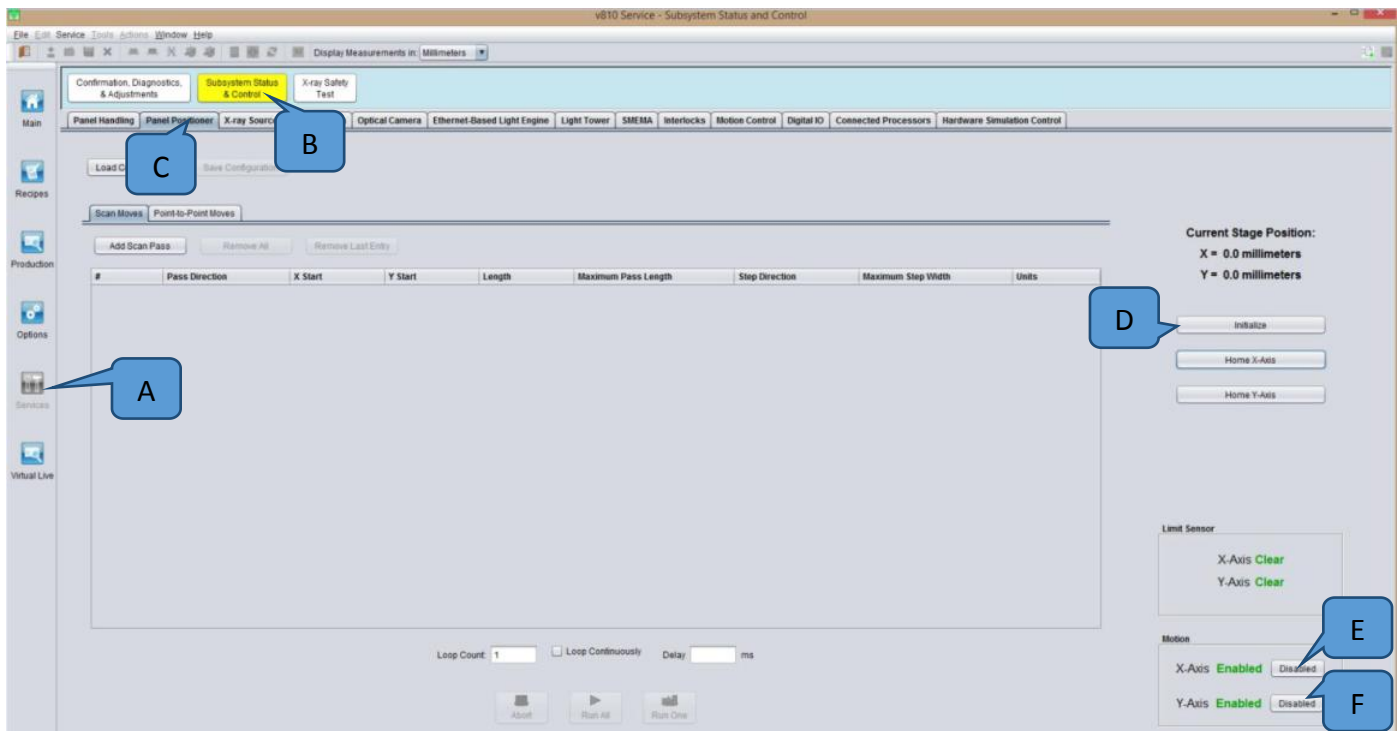


Figure 3 Panel Positioner

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

- Move the Y axis stage towards to Front Access Door until the stage is collide with Y axis minimum hard stop.

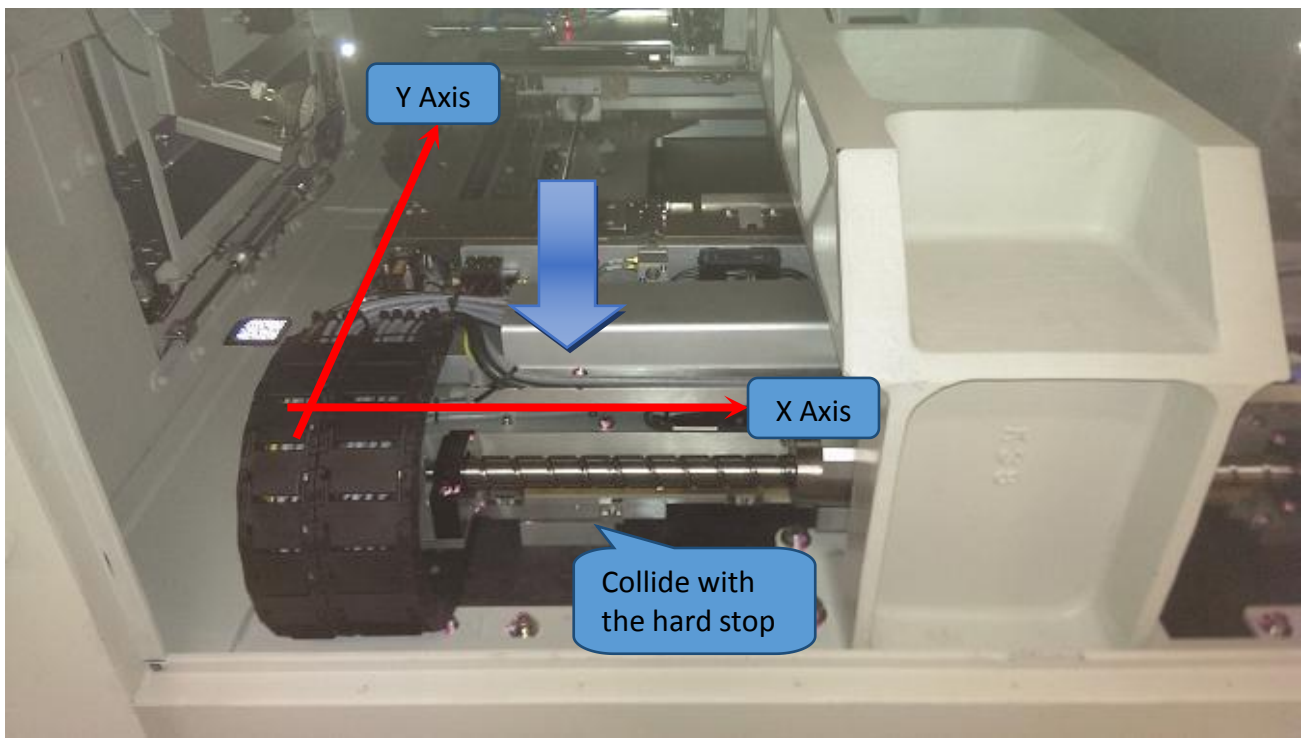


Figure 4 Y axis Minimum Hard Stop

- At Panel Handling tab, trigger Left Outer Barrier to Open Position for easy access.

Services → Subsystem Status & Control → Panel Handling → Open Left Outer Barrier

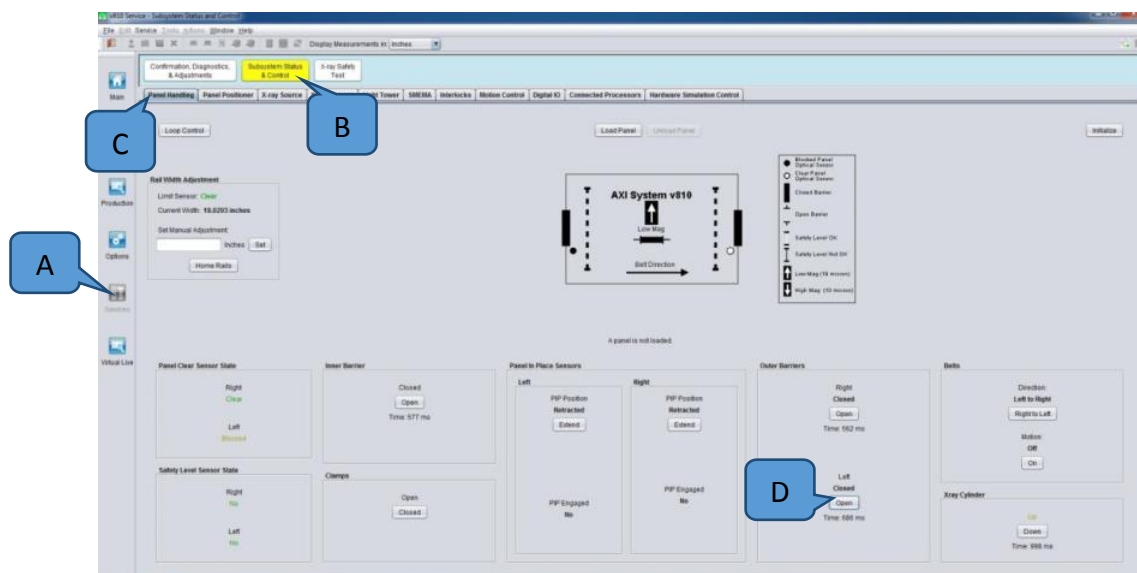


Figure 5 Panel Handling

	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

6. Check on the update value at Current Stage Position (**Panel Handling tab**).

- Spec must within **-3.00 mm ~ -3.30 mm**.
- During **stage Home position**, Limit Sensor show **Clear**, when move to **- 1 mm**, will start show **Block** until the end collide with left hard stop.
- Ensure the Limit Sensor Magnet orientation is correct (Near Machine rear side)
- Ensure the Home Sensor mount at the tap hole near machine rear side

Example#**This will depend to the actual value show at existing machine.**

Y= -3.976mm and need move both sensor toward to machine Rear side approximately 1 mm to optimize min travel stroke.

Red sign assume as starting point.

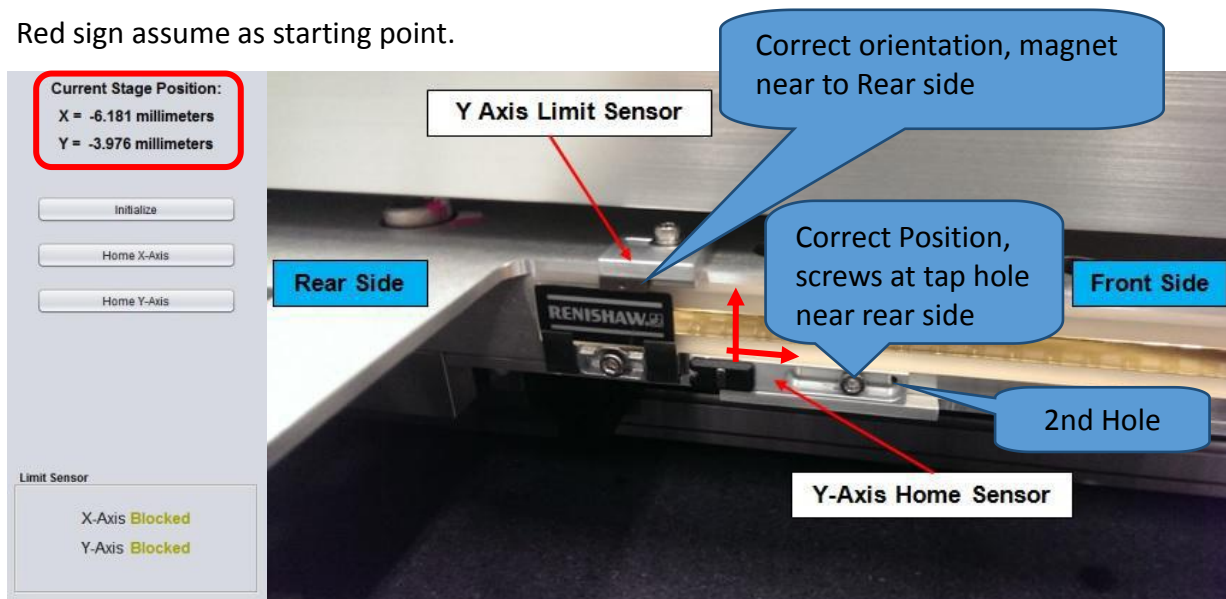


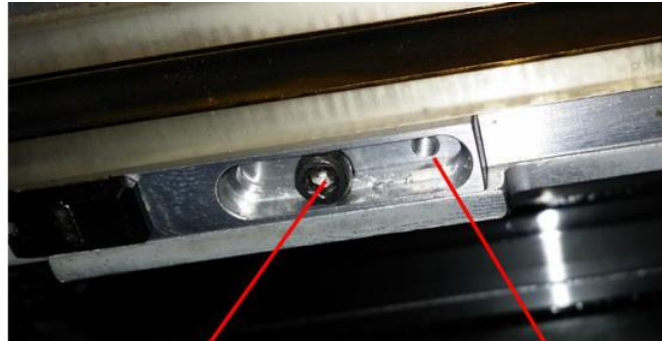
Figure 6 Home Sensor & Limit Sensor

	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

Y Axis Optimization

Home sensor adjustment

Special Remark :



Correct slot of home sensor bracket
(Which near to the rear side of
system)

Wrong slot of home sensor bracket
(Which near to the front side of system)

Figure 7 Home Sensor Slot

7. Gently move the Y axis stage until Y axis Home Sensor bracket is reveal.
8. Loosen the M3 screw as shown below in the Figure 8.

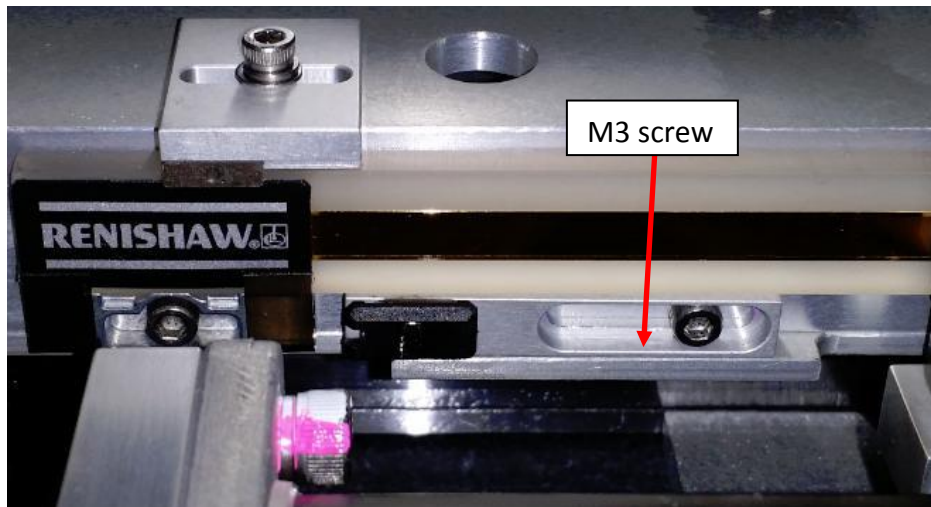


Figure 8 Y-axis Home Sensor M3 Screw

9. Adjust the Y-axis as below table

Current stage position after initialize the Panel Positioner	Direction of the home sensor need to be adjusted
Less than -3.00mm	Towards the front access door
More than -3.30mm	Towards the rear access door

	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

10. Once home sensor adjustment complete, tighten M3 screw as shown in Figure 8.
11. Push the X axis & Y axis stage to the middle of the machine and close front access door.
12. Initialize at Panel Positioner Tab.
13. Open up front access door.
14. Move Y axis stage towards Front Side until the Y axis stage is collide with Y axis Minimum hard stop.
15. Check on the update value at Current Stage Position (Panel Handling tab).

*Spec must within **-3.00 mm ~ -3.30 mm.***

16. Repeat Step 7-15 until Y axis home sensor is within the range from -3.00mm to -3.30mm.
17. In the Panel Positioner tab, change the **Display Measurement** to **Nanometres** and click Initialize.

After the initialization complete, disable the motion for the X-axis and Y-axis.

	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

18. Open the front access door and push the Y axis towards rear direction until it reach Y axis Maximum Hard Stop.

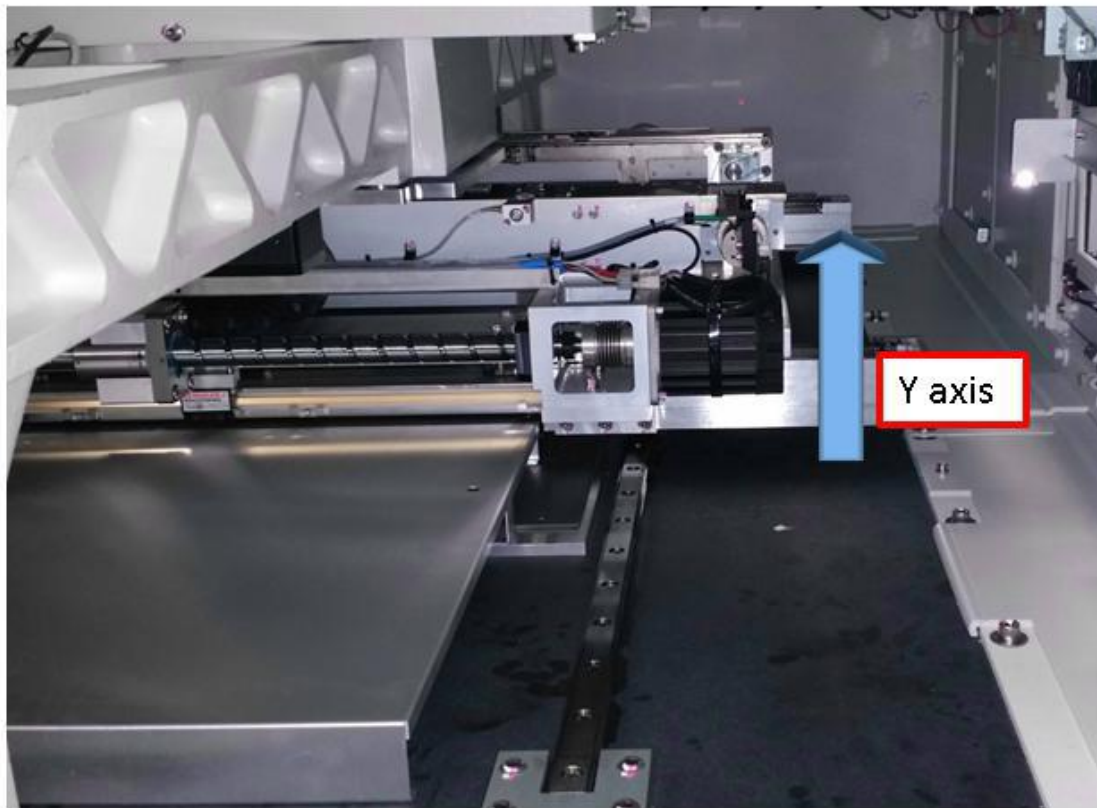


Figure 9 Y axis Maximum Hard Stop

19. Verify Y axis Actual max travel stroke (Display at Panel Positioner `s Current Position) where MUST more than Command Distance by refer to below Machine Model.

Machine Model	Command Distance (NM)	Y axis Actual Max Travel Stroke (NM)
V810 S2	593090000	Required to be more than Command Distance Value
V810 XXL S2	829635920	Required to be more than Command Distance Value
V810 S2EX	660000000	Required to be more than Command Distance Value

	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

Limit sensor adjustment

20. Close the V810 software.

21. Login to the Atom PC and enter the computer name as shown below in Figure 10 (S1 Series) and Figure 11 (S2 Series)

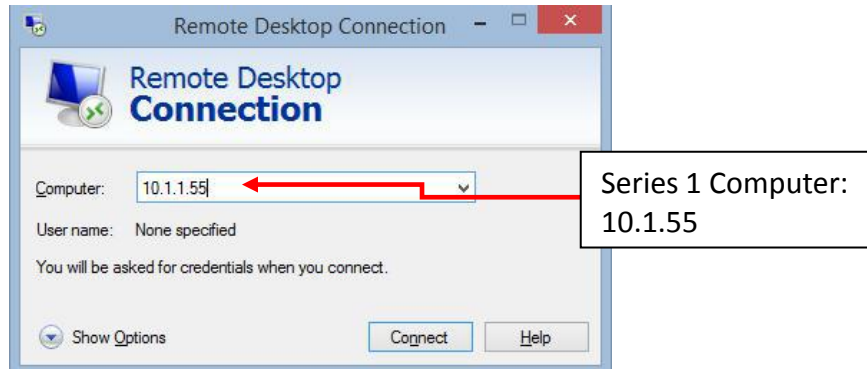


Figure 10 Series 1

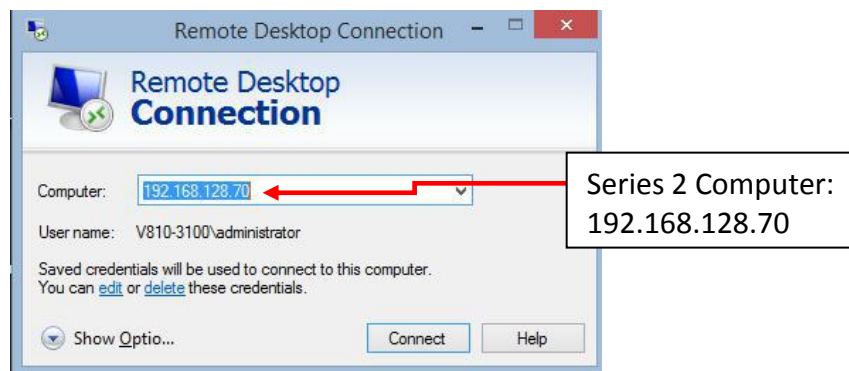


Figure 11 Series 2

22. Enter the password as “Please!” as shown in Figure 12.

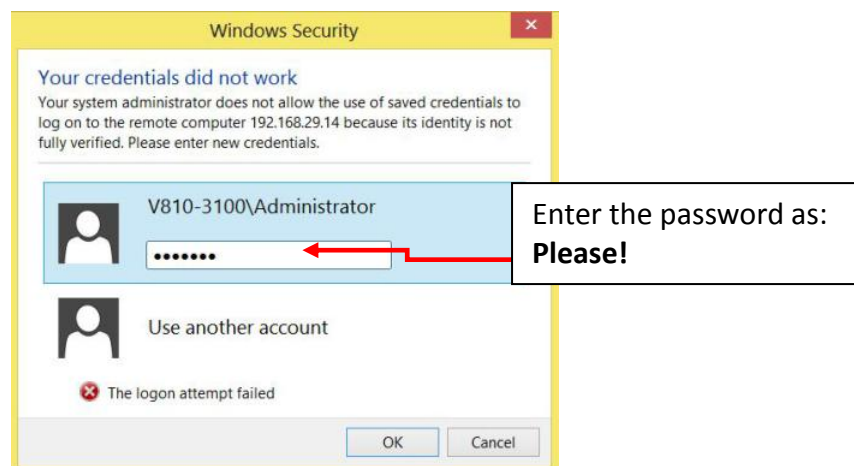


Figure 12 Password for Atom PC

	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

23. Go to the following directory: C:\Program Files\AXI System\v810\REV\bin and open the MC_XMP_NT.EXE as shown in Figure 13.

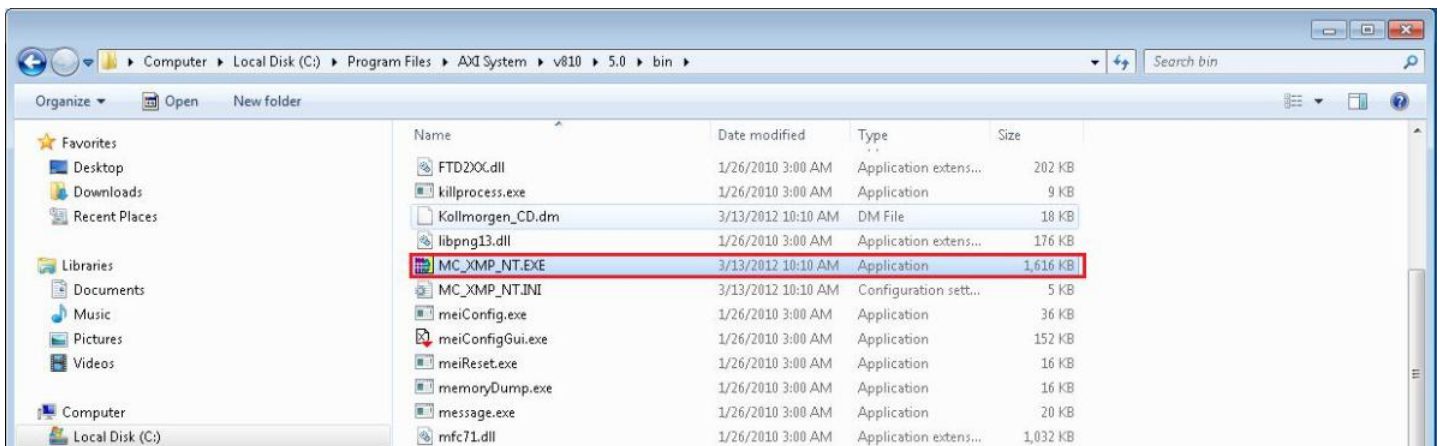


Figure 13 MC_XMP_NT.EXE

24. Ensure this 3 tab is open in the Motion Console tab as shown in Figure 14.

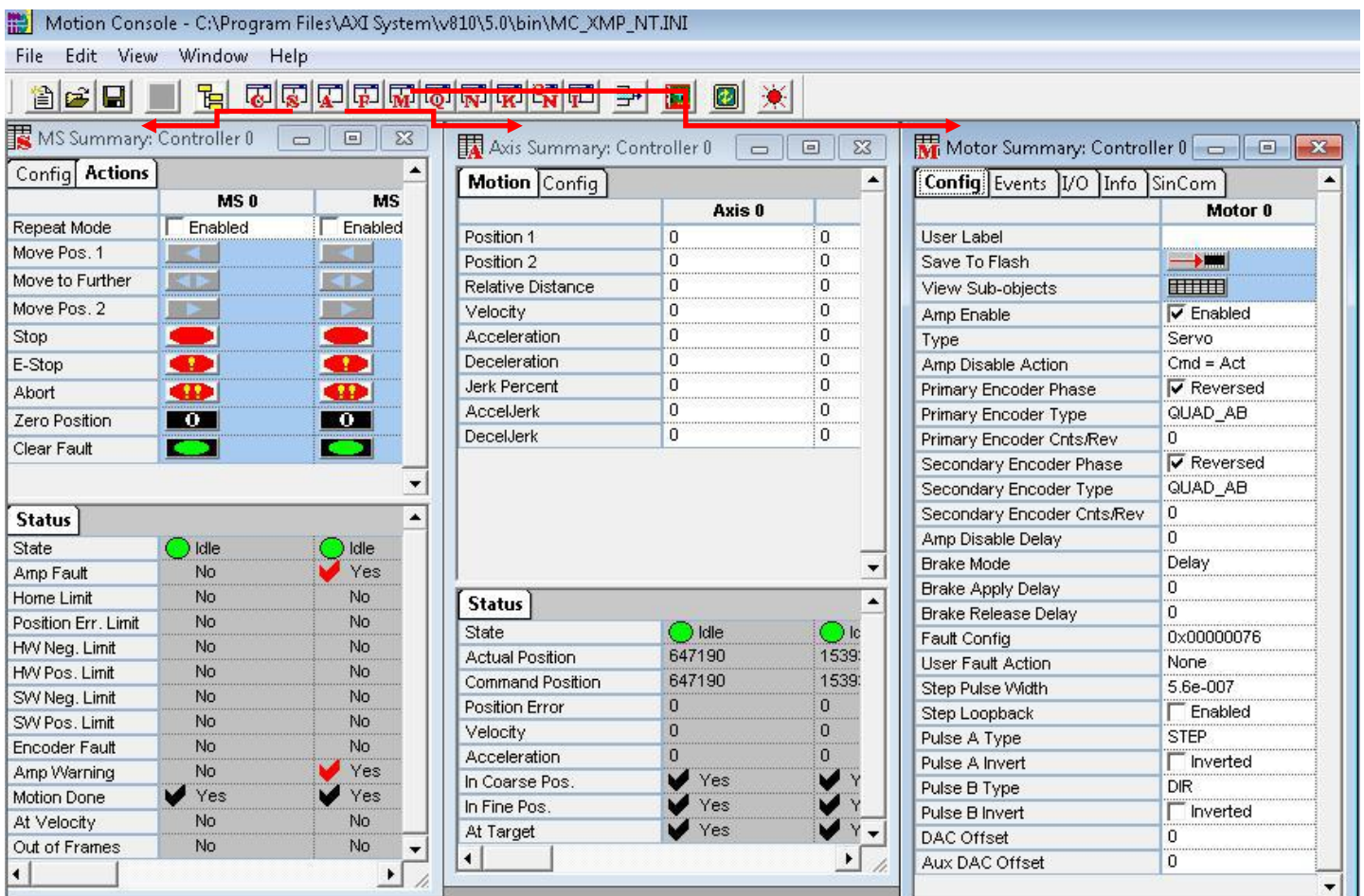


Figure 14 Motion Console tab

ViTrox 	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

25. Close the X-ray source and open the front access door.

26. Push the stage towards the minimum hard stop (front access door) as shown in below Figure 15.

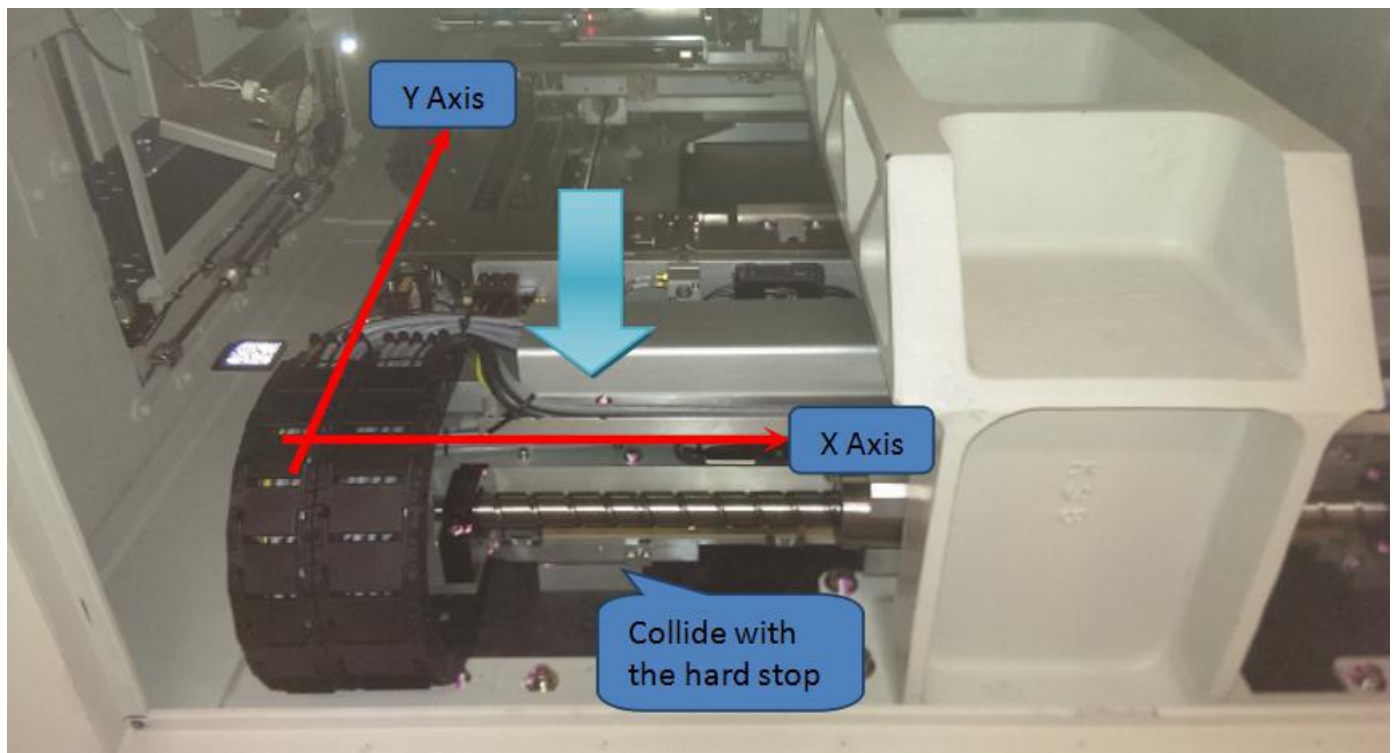


Figure 15 Minimum hard stop

27. Reset all MS0, MS1 and MS2 at the MS Summary (Motion Supervisor Summary) as shown in Figure 16. Once reset for all the axis, the actual position for MS0, MS1 and MS2 will become zero as shown in Figure 17.

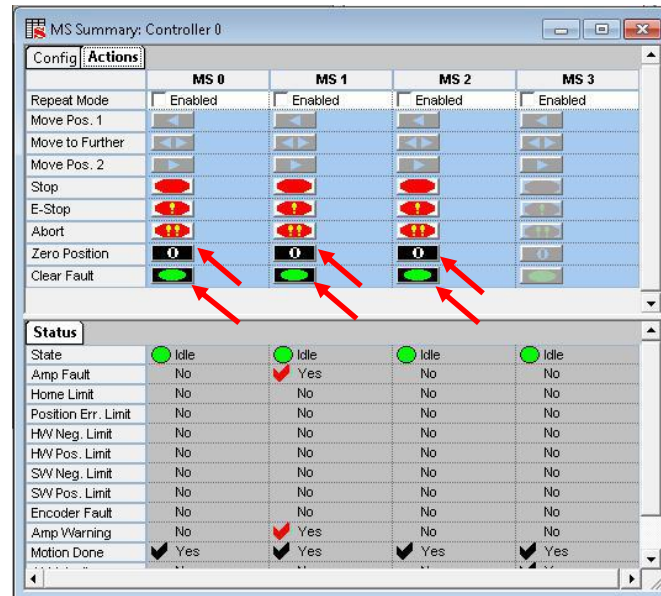


Figure 16 MS Summary (Motion Supervisor Summary)

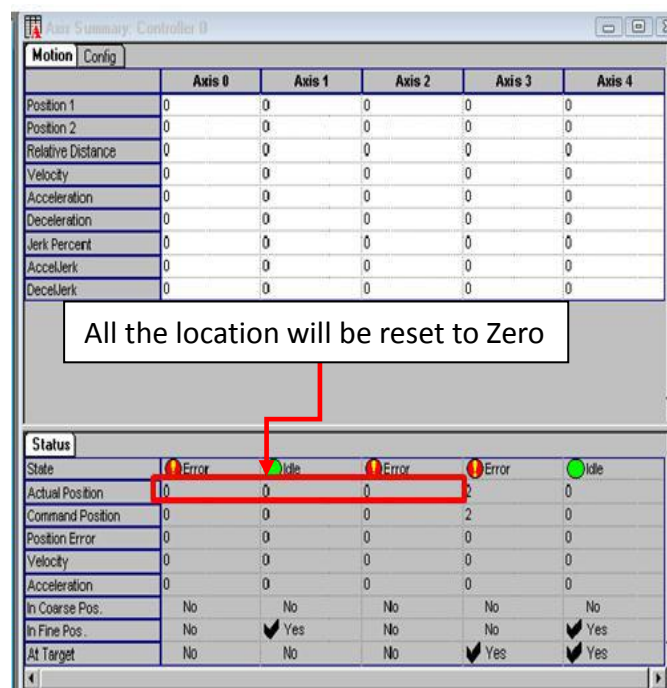


Figure 17 Axis Summary (Actual position)

28. Gently rotate the coupling for the Y-Axis stage as shown in the Figure 18 until the Actual Position and Command Position in Figure 19 become 5990.

Remarks: Acceptable range for the Actual Position and Command Position will be from 5970~5990.

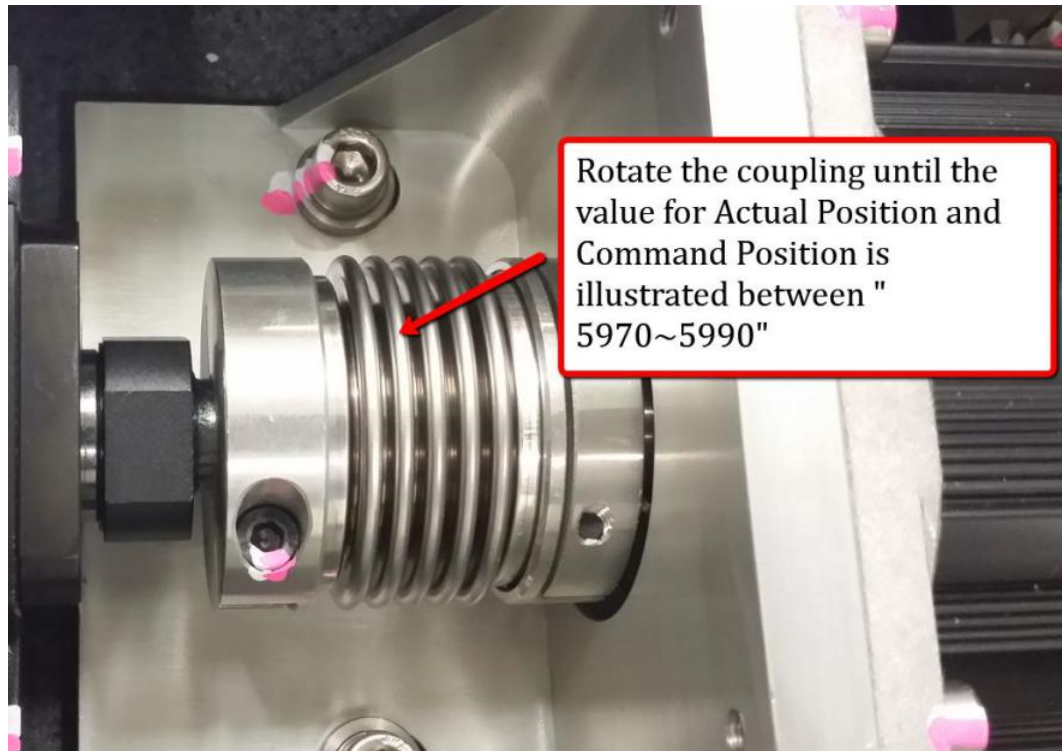
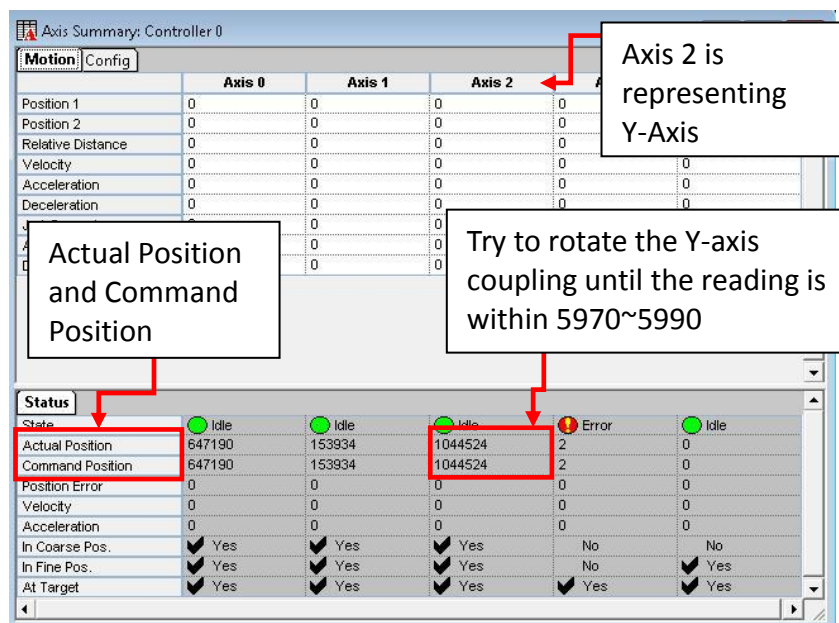


Figure 18 Y-axis coupling



	Axis 0	Axis 1	Axis 2
Position 1	0	0	0
Position 2	0	0	0
Relative Distance	0	0	0
Velocity	0	0	0
Acceleration	0	0	0
Deceleration	0	0	0

	Axis 0	Axis 1	Axis 2
State	Idle	Idle	Idle
Actual Position	647190	153934	1044524
Command Position	647190	153934	1044524
Position Error	0	0	0
Velocity	0	0	0
Acceleration	0	0	0
In Coarse Pos.	Yes	Yes	No
In Fine Pos.	Yes	Yes	No
At Target	Yes	Yes	Yes

Figure 19 Actual Position and Command Position

	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

29. Loosen the M3 screw of the limit switch as shown in Figure 20.

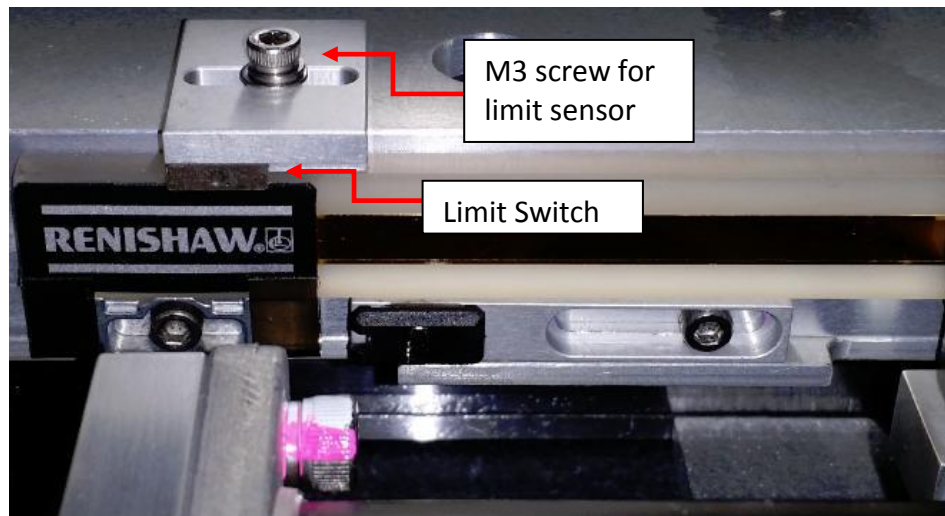
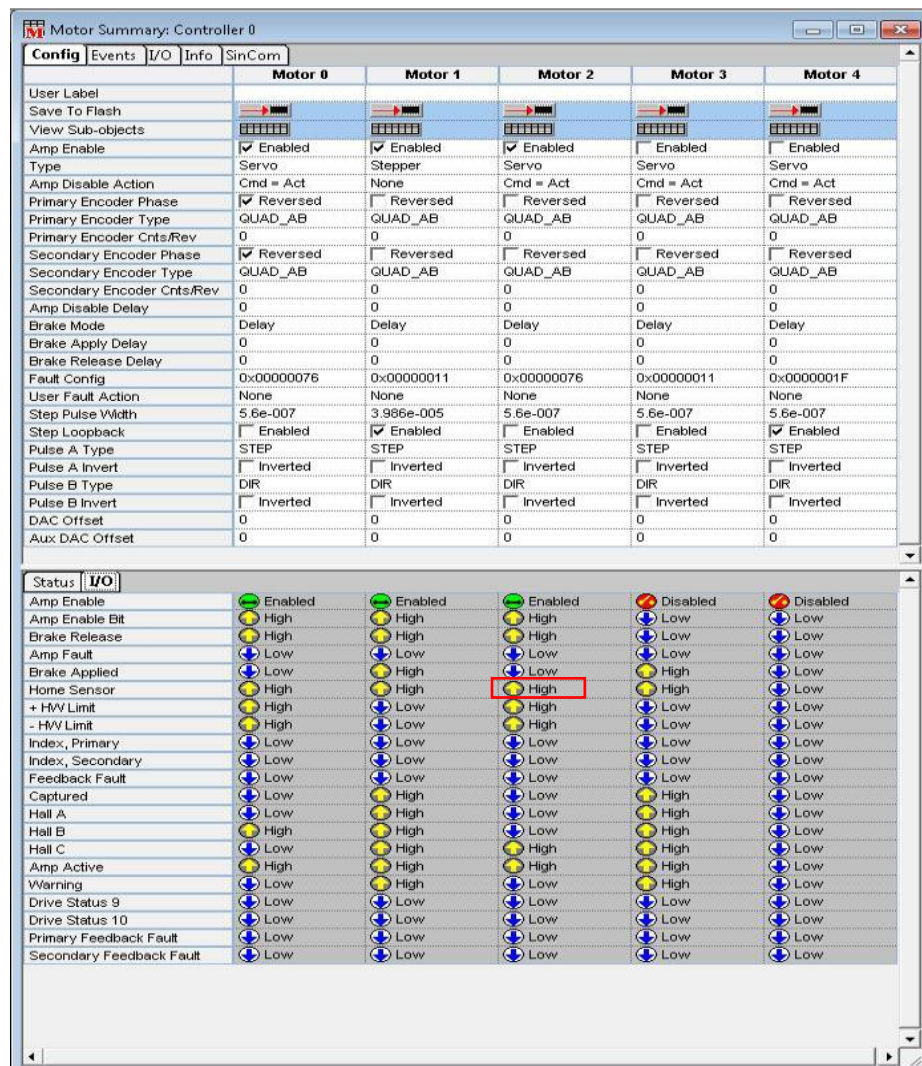


Figure 20 Limit Sensor for Y-Axis

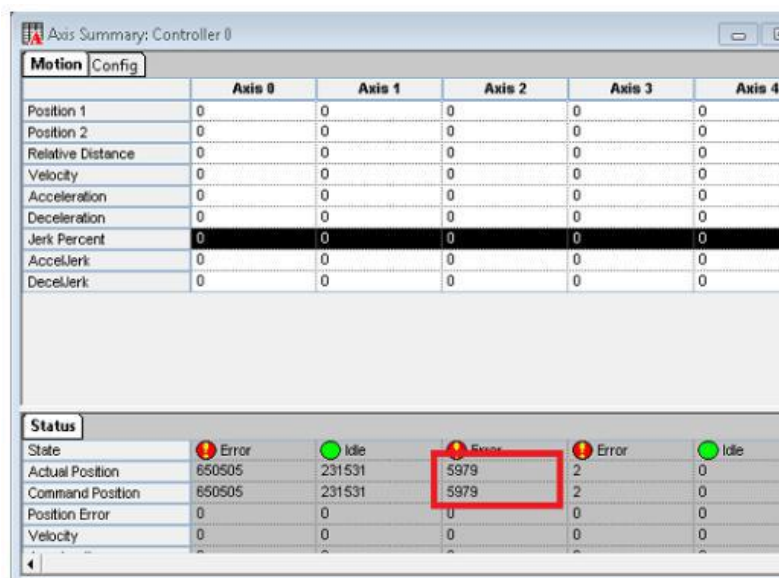
30. Adjust the limit switch shown in Figure 20 until the Home Sensor status become “High” at the Motor Summary as shown in Figure 21 and Figure 22.



	Motor 0	Motor 1	Motor 2	Motor 3	Motor 4
User Label					
Save To Flash					
View Sub-objects					
Amp Enable	☒ Enabled	☒ Enabled	☒ Enabled	☒ Enabled	☒ Enabled
Type	Servo	Stepper	Servo	Servo	Servo
Amp Disable Action	Cmd = Act	None	Cmd = Act	Cmd = Act	Cmd = Act
Primary Encoder Phase	☒ Reversed	☐ Reversed	☐ Reversed	☐ Reversed	☐ Reversed
Primary Encoder Type	QUAD_AB	QUAD_AB	QUAD_AB	QUAD_AB	QUAD_AB
Primary Encoder Cnts/Rev	0	0	0	0	0
Secondary Encoder Phase	☒ Reversed	☐ Reversed	☐ Reversed	☐ Reversed	☐ Reversed
Secondary Encoder Type	QUAD_AB	QUAD_AB	QUAD_AB	QUAD_AB	QUAD_AB
Secondary Encoder Cnts/Rev	0	0	0	0	0
Amp Disable Delay	0	0	0	0	0
Brake Mode	Delay	Delay	Delay	Delay	Delay
Brake Apply Delay	0	0	0	0	0
Brake Release Delay	0	0	0	0	0
Fault Config	0x00000076	0x00000011	0x00000076	0x00000011	0x0000001F
User Fault Action	None	None	None	None	None
Step Pulse Width	5.6e-007	3.986e-005	5.6e-007	5.6e-007	5.6e-007
Step Loopback	☐ Enabled	☒ Enabled	☐ Enabled	☐ Enabled	☒ Enabled
Pulse A Type	STEP	STEP	STEP	STEP	STEP
Pulse A Invert	☐ Inverted	☐ Inverted	☐ Inverted	☐ Inverted	☐ Inverted
Pulse B Type	DIR	DIR	DIR	DIR	DIR
Pulse B Invert	☐ Inverted	☐ Inverted	☐ Inverted	☐ Inverted	☐ Inverted
DAC Offset	0	0	0	0	0
Aux DAC Offset	0	0	0	0	0

	Motor 0	Motor 1	Motor 2	Motor 3	Motor 4
Amp Enable	Enabled	Enabled	Enabled	Disabled	Disabled
Amp Enable Bit	High	High	High	Low	Low
Brake Release	High	High	High	Low	Low
Amp Fault	Low	Low	Low	Low	Low
Brake Applied	Low	High	Low	High	Low
Home Sensor	High	High	High	High	Low
+ H/W Limit	High	Low	High	Low	Low
- H/W Limit	High	Low	High	Low	Low
Index, Primary	Low	Low	Low	Low	Low
Index, Secondary	Low	Low	Low	Low	Low
Feedback Fault	Low	Low	Low	Low	Low
Captured	Low	High	Low	High	Low
Hall A	Low	High	Low	High	Low
Hall B	High	High	Low	High	Low
Hall C	Low	High	High	High	Low
Amp Active	High	High	High	High	Low
Warning	Low	High	Low	High	Low
Drive Status 9	Low	Low	Low	Low	Low
Drive Status 10	Low	Low	Low	Low	Low
Primary Feedback Fault	Low	Low	Low	Low	Low
Secondary Feedback Fault	Low	Low	Low	Low	Low

Figure 21 Motor Summary (Home Sensor)



	Axis 0	Axis 1	Axis 2	Axis 3	Axis 4
Position 1	0	0	0	0	0
Position 2	0	0	0	0	0
Relative Distance	0	0	0	0	0
Velocity	0	0	0	0	0
Acceleration	0	0	0	0	0
Deceleration	0	0	0	0	0
Jerk Percent	0	0	0	0	0
AccelJerk	0	0	0	0	0
DecelJerk	0	0	0	0	0

	Axis 0	Axis 1	Axis 2	Axis 3	Axis 4
State	Error	Idle	Error	Error	Idle
Actual Position	650505	231531	5979	2	0
Command Position	650505	231531	5979	2	0
Position Error	0	0	0	0	0
Velocity	0	0	0	0	0

Figure 22 Axis Summary (Home Sensor)

31. Once the limit switch condition is acquired, tighten back the M3 screw for the limit sensor screw.
32. Pull the Y-Axis stage to the minimum hard stop (front access door).
33. Reset again the Axis 2 become zero as shown in Figure 23 and ensure the actual position for the Axis 0, Axis 1 and Axis 2 become zero as shown in Figure 24.

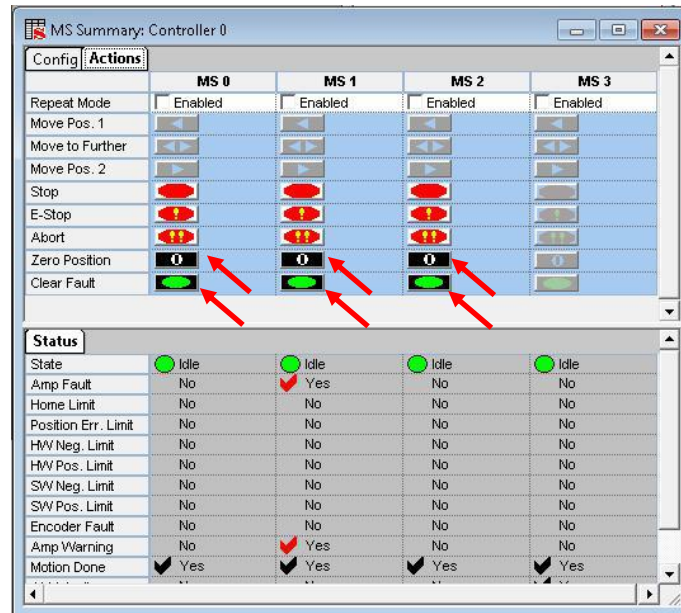


Figure 23 MS Summary

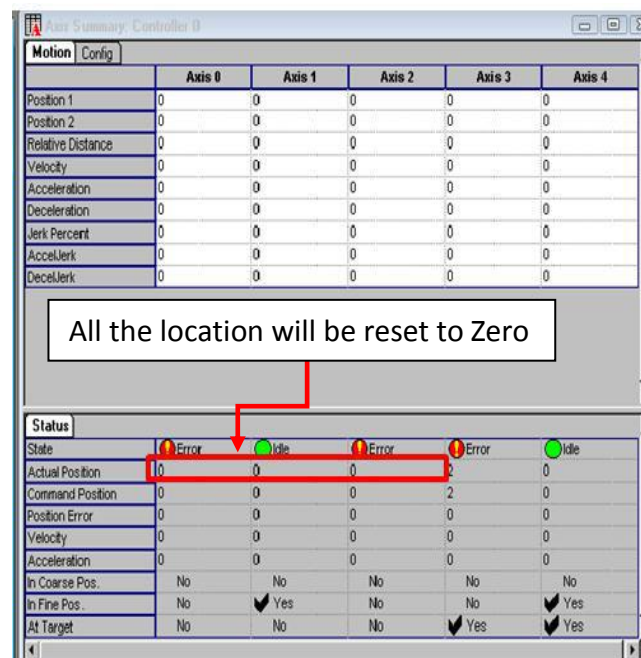


Figure 24 Axis Summary (Actual Position)

34. Slowly rotate the Y-Axis coupling until status become “Low” at indicated at the “Home Sensor” as shown in Figure 16.

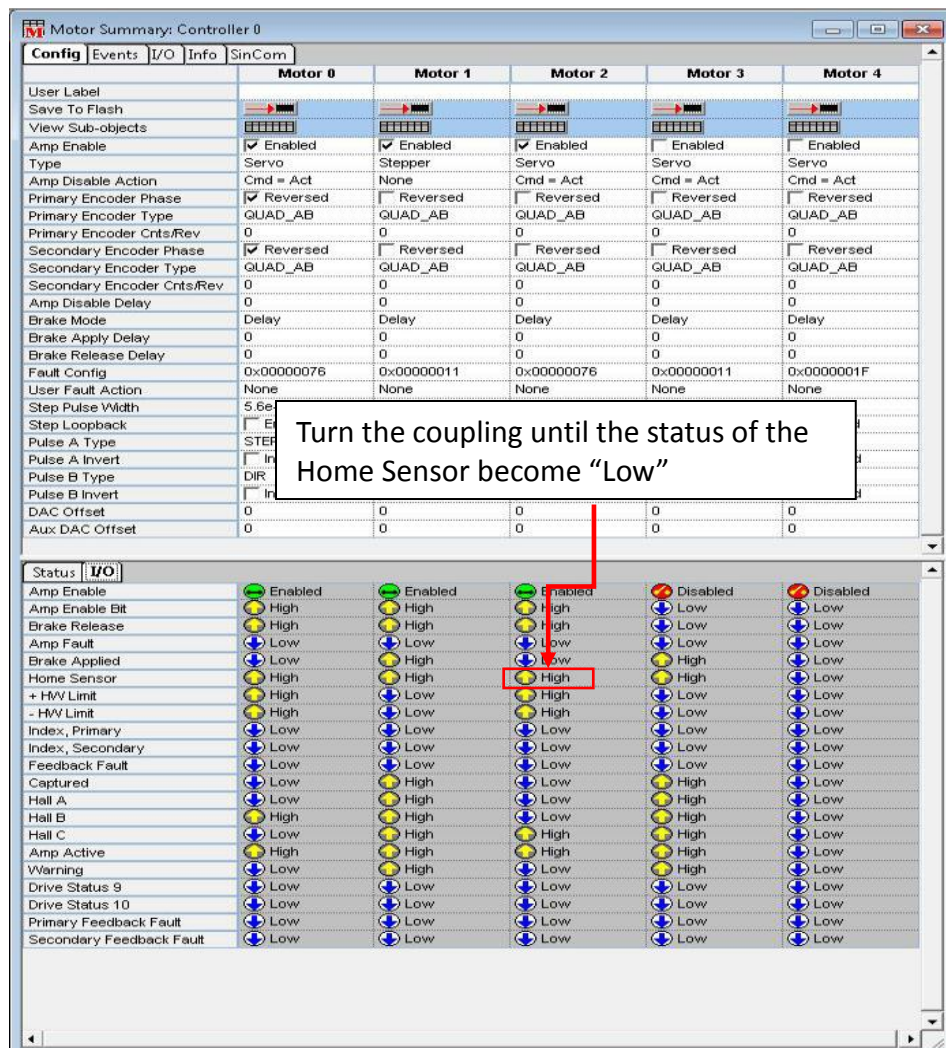


Figure 25 Motor Summary

35. Once the status becomes “Low”, Y-Axis stage should not move anymore.

36. Refer to the actual (From Axis Summary) to verify the limit sensor position. May refer to below table for counter reference.

Y-Axis Actual position (Axis 2 at Axis summary)	Expected result at Motor 2 (Motor summary)
5000~5200	Home Sensor 
5700~5990	Home Sensor 

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

37. Transition signal from low to high will happen at the range of 5000-6000 counts. If the result is not able to achieved as above table (Low at 5000-5200 and High at 5700-5990), please do repeat from Step 24 -36.
38. Exit the Motion Console software.
39. Push the XY-stage to the middle of the machine and close the front access door.
40. Open the V810 software. Go to the Services->Subsystem Status & Control-> Digital IO tab and Click Initialize.
41. Go to the Panel Positioner tab and click Initialize. The limit sensor status for the Y-axis should be clear. Disable the Motion for the X-axis and Y-axis as shown in Figure 26. Close the X-ray source and open the front door.

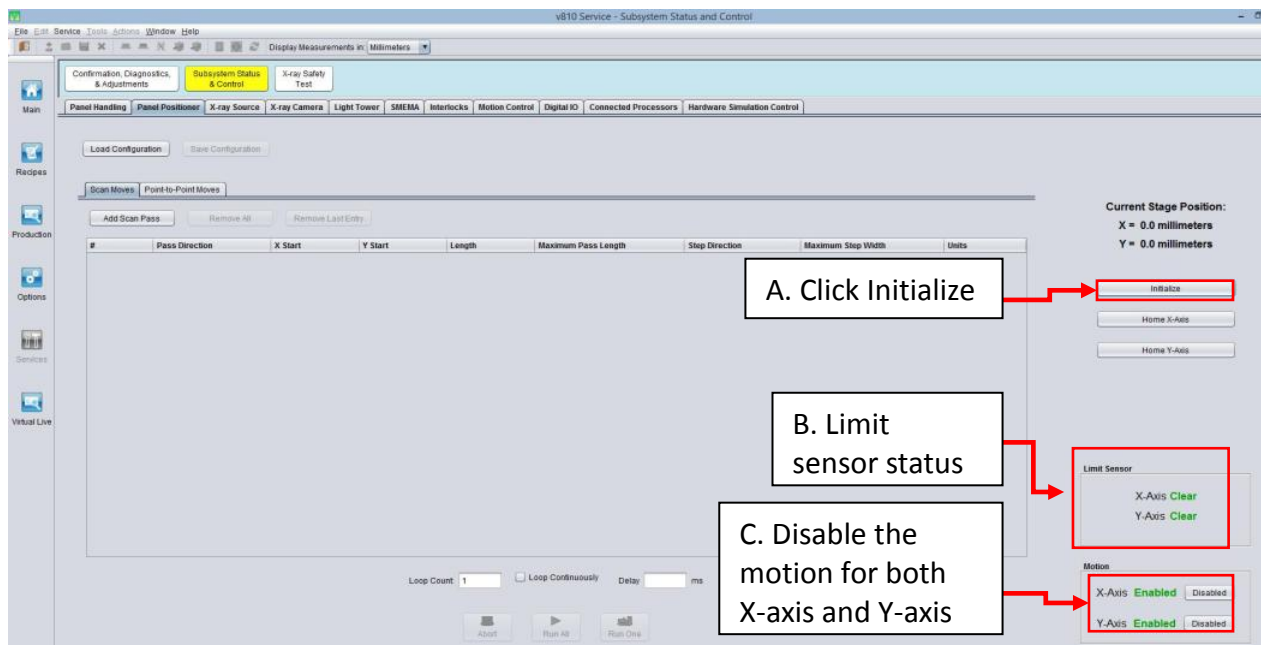


Figure 26 Panel Positioner

	Work Instruction	Doc. Revision	01
	AXI V810 Y-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

42. Push the stage to the minimum hard stop (left outer barrier side) and the status of the X-axis limit sensor should be blocked as shown in Figure 27.

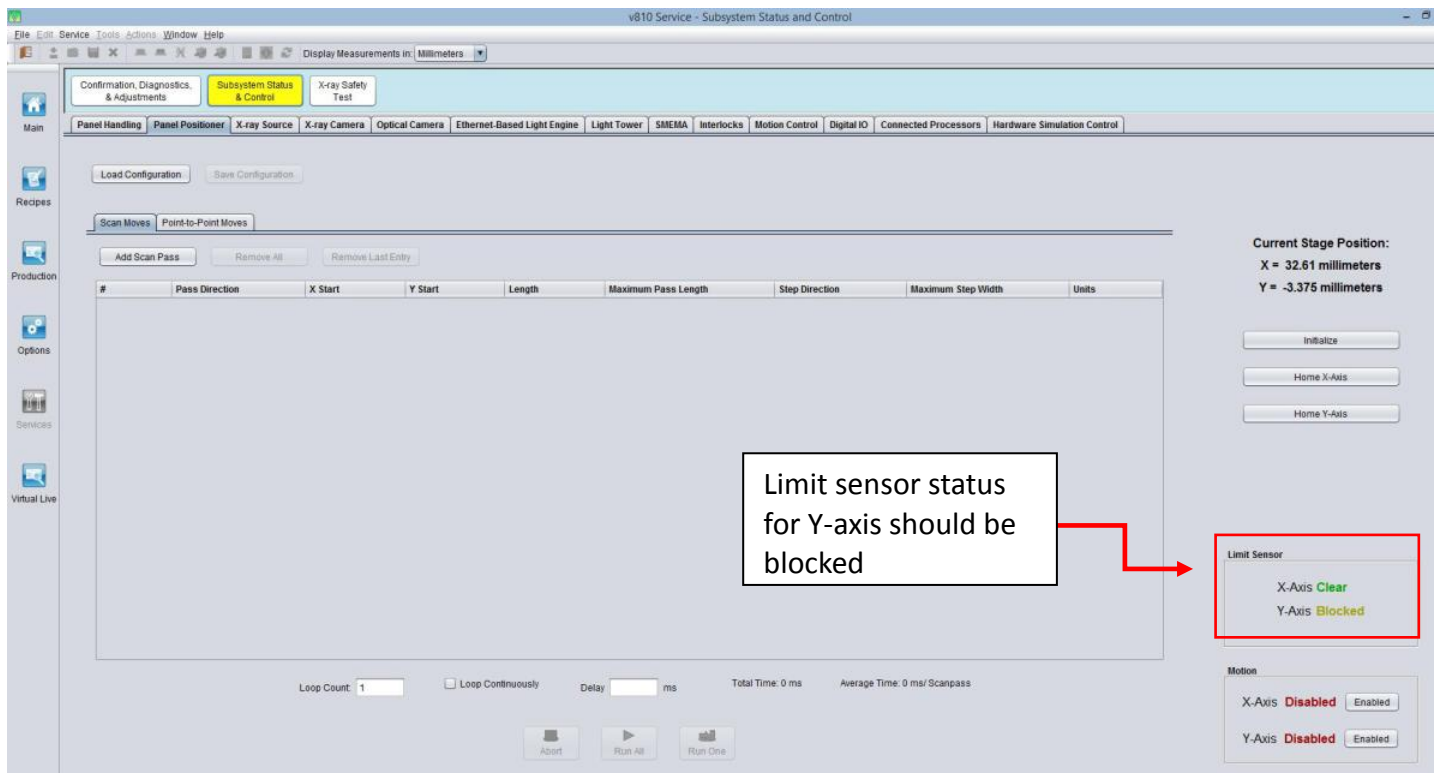


Figure 27 Y axis Limit Sensor

43. If verification not be able to achieved, repeat Step 24-42.